2-Dars

Model Evaluation

- Confusion Matrix:
  - Table → jadval
  - Actual vs Predicted

bu actual / predict qiymatlarni orasi bilan chaydi

- Metrics:
  - Accuracy
  - Precision
  - Recall
  - F1-Score
  - Support

Metrics jamlanmasi:

Classification_report

Confusion matrix

y_actual	y_pred	Sample
1	1	1
0	0	2
1	0	3
0	0	4
1	1	5

	predicted 0	predicted 1
Actual 0	True Negative	False positive
Actual 1	False Negative	True positive

0 -> Negative
1 -> positive

	pred 0	pred 1
A 0	2	0
A 1	1	2

Cod:

```
from sklearn.metrics import
confusion_metrix, classification=>
=> _report
```

```
y_actual = [1, 0, 1, 0, 1]
y_pred = [1, 0, 0, 0, 1]
cm = confusion_metrix(y_actual, =>
=> y_pred)
```

cm -> call

```
array([[2, 0],
       [1, 2]], dtype=int64)
```

Accuracy

$$Acc = \frac{TP+TN}{TP+FP+TN+FN}$$

$$Acc = \frac{2+2}{5} = \frac{4}{5} = 0,8 \cdot 100 = 80\%$$

ЧИСЛО МЕСЯЦ ГОД ПН ВТ СР ЧТ ПТ СБ ВС

Precision

positive chiqganlarni % ni hisoblab beradi:

$$precision = \frac{TP}{TP+FP} = \frac{2}{2+0} = \frac{2}{2} = 1.$$

positive

Recall

$$\frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} \approx 0,67\%$$

F1-Score

recall / Precision

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ЧИСЛО МЕСЯЦ ГОД ПН ВТ СР ЧТ ПТ СБ ВС

$$F1 = \frac{2pre \cdot recall}{pre + recall} = 2 \cdot \frac{pre \cdot re}{pre + re} =$$

$$= 2 \cdot \frac{1 \cdot 0,67}{1 + 0,67} \approx 0,8$$

Support = class larni teng tagsin qilmaganini bilib beradi.

Linear Models: Logistic Regression

- Logistic Regression → sigmoid function (logic function)

$$\delta = \frac{1}{1+e^{-z}}$$

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